



Technical Service Bulletin

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Modular Common Rail System (MCRS) Fuel Injector Clamp Chamfer Cutting Tool

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Warranty Statement

The information in this document has no effect on present warranty coverage or repair practices, nor does it authorize TRP or Campaign actions.

Contents

Product Affected

- QSK19 CM850 MCRS
- QSK19 CM2150 MCRS
- QSK19 CM2350 MCRS K105
- QSK19 CM2350 MCRS K114
- QSK19 CM2350 MCRS K127
- QSK38 CM850 MCRS
- QSK38 CM2150 MCRS K106
- QSK38 CM2350 MCRS K125
- QSK38 CM2350 MCRS K138M
- QSK50 CM850 MCRS
- QSK50 CM2150 MCRS
- QSK50 CM2150 MCRS K107
- QSK50 CM2350 MCRS K108
- QSK50 CM2350 MCRS K130
- QSK50 CM2350 MCRS K146G
- QSK60 CM850 MCRS
- QSK60 CM2150 MCRS
- QSK60 CM2350 MCRS K116
- QSK60 CM2350 MCRS K117
- QSK60 CM2350 MCRS K121
- QSK60 CM2350 MCRS K135
- QSK60 CM2350 MCRS K136
- QSK60 CM2350 MCRS K149M
- QSK78 CM2350 MCRS K126
- QSK78 CM2350 MCRS K137

Issue

Symptom:

- Injector clamp capscrew malfunction allowing injector protrusion from the valve cover.

Root Cause:

- Interference between the fuel injector clamp chamfer and capscrew.

Verification

The fuel injector clamp chamfer cutting tool is recommended for use on any High Horsepower Modular Common Rail System (MCRS) engine where injectors are being serviced requiring a new capscrew to be installed. If interference between the capscrew fillet and clamp chamfer is identified (visual confirmation of a gap between the capscrew head and the clamp top surface while **not** installed in the engine, or by any measured gap between the capscrew head and clamp top surface in an installed condition), the injector clamp **must** have the chamfer modified through use of the chamfer cutting tool. If the issue was identified in the installed condition, the injector hold down clamp capscrew **must** be replaced on re-installation.

Resolution

A new injector clamp chamfer cutting tool, Part Number 5395114, was released to provide a controlled method for reshaping the capscrew bore chamfer on the injector hold-down clamp. The tool features a bore guide and a depth-controlling shoulder and is designed for use with powered hand tools.

A revised injector clamp has been released that incorporates a modified chamfer design to accommodate the dimensional tolerance of the existing capscrew. See Table 1 below for injector clamp service parts information. The chamfer cutting tool dimensions reflect the same dimensions as the revised injector clamp.

Service Parts Availability

Existing clamps already in service can be replaced with the revised injector clamp or they can be modified with the chamfer cutting tool during service events.

Revised clamp clean date is October 1, 2020.

Prior to plant implementation of the revised injector clamp, modified supplier parts under the same part number as the existing clamp were provided to plants to support manufacturing. The build period for these modified parts began on 16 March 2017 and extended through the implementation dates of the new parts listed below in Table 2.

Service parts available. See Table 1 below for part numbers.

Table 1, Modular Common Rail System (MCRS) Injector Clamp Service Parts

Part Description	Existing Part Number	Obsolete	Superseded	New Part Number
CLAMP,INJECTOR	4917436	Yes	Yes	5375028

Table 1, Modular Common Rail System (MCRS) Injector Clamp Service Parts				
Part Description	Existing Part Number	Obsolete	Superseded	New Part Number
TOOL,CHAMFER CUTTING	-	-	-	5395114
Production Status				
Implemented for production. See Table 2				
Table 4, Production Information				
ESN First	Build Date ¹		Plant	
33214154	08 May 2018		DAV-Daventry Engine Plant	
37274959	04 May 2018		CIC/SEP – Seymour Engine Plant	
¹ Engine build date can be found on the engine data plate.				

Document History

Date	Details
2019-1-9	Module Created
2019-1-29	Quick Fix application:
2020-11-16	Non-Product Problem Solving (PPS)

Last Modified: 02-Dec-2020
